Ren *et al. BMC Public Health* (2025) 25:3638
https://doi.org/10.1186/s12889-025-24807-5

BMC Public Health

SYSTEMATIC REVIEW Open Access

Preventive effects of different types of exercises during pregnancy on gestational diabetes: a systematic review and network meta-analysis

Yijia Ren[1,2†], Liuwei Zhang[1,2*†], Jiaoying Cheng[3*], Yi Gao[1,4] and Yuqing He[1,5]

Abstract

Background Exercise is known to reduce the risk of gestational diabetes mellitus (GDM). However, the effectiveness of various exercise regimens in preventing GDM remains unclear, and the optimal approach has yet to be identified. This study aims to perform a systematic review and network meta-analysis to evaluate the preventive effects of different exercise interventions on GDM and to identify the most effective exercise strategy for its prevention.

Methods A comprehensive search of seven databases was conducted for studies published before March 2025 that examined the relationship between exercise during pregnancy and the risk of GDM. All literature underwent a rigorous screening process based on pre-defined inclusion and exclusion criteria. The quality of evidence was assessed using the Cochrane Risk of Bias (RoB) tool. Data analysis, heterogeneity testing, publication bias assessment, and sensitivity analysis were performed using the Stata 17.0 software.

Results (1) Meta-analysis revealed that the risk of GDM was significantly lower in the exercise intervention group compared to the conventional care group (OR 0.56, 95% CI = 0.44–0.71). (2) Subgroup analysis indicated that aerobic exercise (OR 0.40, 95% CI = 0.25–0.64), yoga (OR 0.19, 95% CI = 0.05–0.69), and diet and exercise guidance (OR 0.53, 95% CI = 0.37–0.76) significantly reduced the risk of GDM compared to the control group. Exercise intervention during pregnancy effectively reduced the risk of GDM in healthy individuals (OR 0.63, 95% CI = 0.43–0.93), overweight and/or obese individuals (OR 0.53, 95% CI = 0.36–0.79), and pregnant women at risk of GDM (OR 0.53, 95% CI = 0.34–0.82). Additionally, exercise intervention initiated in early pregnancy significantly prevented the development of GDM (OR 0.51, 95% CI = 0.39–0.68). (3) Network meta-analysis results demonstrated that yoga had the lowest surface under the cumulative ranking curve (SUCRA = 6.8%), suggesting that yoga was the most effective intervention in reducing the risk of GDM.

†Yijia Ren and Liuwei Zhang contributed equally and should be considered co-first authors.

*Correspondence:
Liuwei Zhang
2466@bsu.edu.cn
Jiaoying Cheng
cjy19781031@163.com

Full list of author information is available at the end of the article

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Conclusion Various types of exercise, including yoga, aerobic exercise, and diet and exercise guidance, are effective in preventing GDM during pregnancy. Among these, yoga appears to be particularly promising and could be considered for inclusion in preventive care strategies for pregnant women.

Trial registration The study was registered on the International Prospective Registry of Systematic Reviews (PROSPERO) database (CRD42024552641) on 10 June 2024.

Keywords Exercise, Yoga, Gestational diabetes mellitus, Network meta-analysis

Introduction

Gestational diabetes mellitus (GDM) is defined as glucose intolerance first identified during pregnancy and is one of the most prevalent complications associated with pregnancy [1]. Immediate consequences of GDM for mothers include an increased risk of gestational hypertension and polyhydramnios, while for infants, it raises the likelihood of fetal macrosomia, shoulder dystocia, and neonatal hyperbilirubinemia. In the long term, GDM poses significant health risks for both mothers and children, including a heightened likelihood of developing type 2 diabetes mellitus (T2DM) and metabolic syndrome in the postnatal periods [2]. According to the International Association of Diabetes and Pregnancy Study Groups (IADPSG), the global prevalence of GDM is estimated to be 14.0% [3].

Exercise intervention is considered a key component in the management of GDM [4]. The 2020 World Health Organization (WHO) Physical Activity Guidelines [5] recommend that pregnant women without contraindications engage in at least 150 min of moderate-intensity aerobic activity per week, alongside daily physical activity. The dose–response relationship between physical activity and GDM risk was demonstrated in our prior systematic review, which indicated that a physical activity volume of approximately 50 MET h/week was associated with the greatest reduction in GDM risk [6]. Although this threshold is somewhat higher than the minimum level recommended by the WHO guidelines, it highlights the potential additional benefits of engaging in physical activity beyond the minimum recommendation. In addition, the WHO guidelines encourage a variety of exercise modalities during pregnancy, including aerobic exercise, muscle-strengthening activities, and gentle stretching. These forms of exercise have shown significant benefits for both maternal and fetal health. For pregnant women, exercise is an effective strategy for preventing various pregnancy-related complications, including GDM, excessive gestational weight gain, hypertension, urinary incontinence, lumbar and pelvic pain, anxiety, and prenatal depression [7]. Moreover, exercise during pregnancy has been associated with reduced risk of preterm birth, neonatal complications, and postpartum depression [8].

Despite the guidelines recommending exercise during pregnancy [5], several issues require further investigation. Firstly, there is a lack of consensus on the efficacy of different exercise types in preventing GDM. While some evidence indicates that yoga [9, 10], aerobic exercise [11], aerobic combined with resistance exercise [12], and diet and exercise guidance [13, 14] can effectively prevent GDM, other studies have reported inconsistent findings [15–17]. Secondly, prior research has primarily focused on managing and controlling blood glucose levels through exercise interventions, with limited attention to the potential preventive effects of early exercise on GDM. Early and proactive exercise interventions play a critical role in promoting maternal health during pregnancy, not only by preventing GDM but also by providing broader health benefits for both mothers and infants [18]. Thirdly, most studies have compared single exercise types with conventional care, with limited comprehensive comparisons among various exercise modalities. As a result, there is insufficient evidence to recommend the most suitable exercise strategies for optimal prevention. Thus, this study aims to identify the most effective exercise intervention for preventing GDM through a network meta-analysis. This approach will provide more precise exercise recommendations for pregnant women, helping to optimize prevention strategies for GDM.

Methods

All meta-analyses conducted in this study were conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (see Additional file 1) [19]. The study was registered in the International Prospective Register of Systematic Reviews (PROSPERO) database (CRD42024552641) on 10 June 2024 (available from: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42024552641).

Literature search

A comprehensive search was conducted across multiple electronic databases, including CNKI, Wanfang, Web of Science, PubMed, Cochrane Library, Embase, and EBSCO, to identify relevant literature on the relationship between exercise during pregnancy and the development of GDM. The search utilized the following formula (see Table 1): "pregnancy" OR "pregnant women" OR "maternal" OR "gestation", "gestational diabetes mellitus" OR "gestational diabetes" OR "GDM" OR "diabetic,

Table 1 Systematic literature search terms and strategies

Search terms for PubMed
#1 (pregnancy[Title/Abstract] OR pregnant women[Title/Abstract] OR maternal[Title/Abstract] OR gestation[Title/Abstract])
#2 (Gestational Diabetes Mellitus[Title/Abstract] OR gestational diabetes[Title/Abstract] OR GDM[Title/Abstract] OR diabetic, pregnancy[Title/Abstract] OR Blood glucose[Title/Abstract])
#3 (exercise[Title/Abstract] OR physical activity[Title/Abstract] OR sport[Title/Abstract] OR yoga[Title/Abstract] OR exercise intervention[Title/Abstract])
#1 AND #2 AND #3
Search terms for EBSCO
#1 SU=(pregnancy OR pregnant women OR maternal OR gestation)
#2 SU=(Gestational Diabetes Mellitus OR gestational diabetes OR GDM OR diabetic, pregnancy OR Blood glucose)
#3 SU=(exercise OR physical activity OR sport OR yoga OR exercise intervention)
#1 AND #2 AND #3
Search terms for Web of Science
#1 TI=(pregnancy or gestation or pregnant women or maternal)
#2 TI=(Gestational Diabetes Mellitus or gestational diabetes or GDM or diabetic, pregnancy or Blood glucose)
#3 TI=(exercise or physical activity or sport or yoga or exercise intervention)
#1 AND #2 AND #3
Search terms for Cochrane
#1 Title Abstract Keyword (pregnancy OR gestation OR pregnant women OR maternal)
#2 Title Abstract Keyword (Gestational Diabetes Mellitus OR gestational diabetes OR GDM OR diabetic, pregnancy OR Blood glucose)
#3 Title Abstract Keyword (exercise OR physical activity OR sport OR yoga OR exercise intervention)
#1 AND #2 AND #3
Search terms of Embase
#1 TS=(pregnancy or gestation or pregnant women or maternal)
#2 TS=(Gestational Diabetes Mellitus or gestational diabetes or GDM or diabetic, pregnancy or Blood glucose)
#3 TS=(exercise or physical activity or sport or yoga or exercise intervention)
#1 AND #2 AND #3
Search terms for CNKI (in Chinese)
#1 Title Abstract Keyword = pregnancy + pregnant women + maternal + gestation
#2 Title Abstract Keyword = Gestational Diabetes Mellitus + gestational diabetes + GDM + diabetic, pregnancy + Blood glucose
#3 Title Abstract Keyword = exercise + physical activity + sport + yoga + exercise intervention
#1 AND #2 AND #3
Search terms for Wanfang (in Chinese)
#1 (Title or Keywords: ((“pregnancy”)or(“pregnant women”) or(“maternal”)or(“gestation”)))
#2 (Title or Keywords: ((“Gestational Diabetes Mellitus”)or(“gestational diabetes”)or(“GDM”)or(“diabetic, pregnancy”)or(“Blood glucose”)))
#3 (Title or Keywords: ((“exercise”) or (“physical activity”)or(“sport”) or(“yoga”)or(“exercise intervention”)))
#1 AND #2 AND #3

pregnancy” OR “blood glucose”, and “exercise” OR “physical activity” OR “sport” OR “yoga” OR “exercise intervention”. The search covered studies from the inception of each database up to 31 March 2025. The search was restricted to articles published in Chinese or English.

Eligibility criteria

Inclusion criteria: (1) The study was a randomized controlled trial (RCT) investigating the prevention of GDM through exercise; (2) The study was published in Chinese or English by 30 June 2024; (3) The intervention involved exercise, while the control group received conventional care. (4) The primary outcome measure was the diagnostic results of GDM.

Exclusion criteria: (1) Conference papers, reviews, meta-analyses, animal studies, empirical reports, non-research literature, or other irrelevant publications. (2) Studies that did not include GDM as an outcome indicator or had incomplete data. (3) Chinese articles published in non-core journals. (4) Duplicate publications.

The intervention involved exercise during pregnancy. We categorized the exercise interventions into 6 types: aerobic exercise, resistance exercise, aerobic combined with resistance exercise, yoga, exercise guidance, and diet and exercise guidance. The control group received conventional care. Detailed definitions of these categories are provided in Table 2.

Data extraction

The data extraction was conducted by two researchers (Y.R. and Y.G.) utilizing standardized electronic data extraction tables. In cases of disagreement, a third author (L.Z.) was consulted to resolve conflicts. In the absence of data in the original article, the research authors are contacted via email to request additional information. Extracted data items included the first author, publication year, research population, sample size, intervention details (types of interventions and their start time) for experimental and control groups, and the outcome measures (e.g., diagnosis of GDM).

Quality assessment

The risk of bias (RoB) was evaluated by two independent reviewers (Y.R. and Y.G.) using the Cochrane RoB tool, version 2. This tool evaluates five key domains: randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. For each domain, the reviewers categorized the risk of bias was categorized as “low risk”, “unclear risk”, or “high risk”. Any discrepancies arising during the quality assessment were resolved by the third author (L.Z.).

Table 2 Definitions of intervention and control groups

Intervention group	aerobic exercise	Primarily relying on aerobic metabolism to supply the energy needed for exercise. Examples include cycling, running, aerobic dancing, and other moderate-intensity activities.
	resistance exercise	The active movement of muscles to overcome external resistance, which can be provided by another person, the individual's healthy limbs, or equipment like dumbbells, springs, or resistance bands.
	aerobic combined with resistance exercise	A comprehensive exercise mode combining aerobic and resistance exercise
	yoga	A movement practice aimed at achieving harmony and unity of the body, mind, and spirit, It incorporates methods for adjusting body posture, regulating breathing, and practicing meditation to calm the mind.
	exercise guidance	Recommendations focus on total exercise volume, intensity, and duration, without prescribing a specific exercise mode.
	diet and exercise guidance	Diet and exercise guidance or planning is provided without centralized organizational supervision. Implementation is monitored through methods such as questionnaires, follow-up visits, WeChat communication, and record-keeping.
Control group	conventional care	This refers to the standard prenatal examinations during pregnancy without additional exercise interventions. Some hospitals also provide educational and advisory services focused on maternal health during the perinatal period.

Statistical analysis

Data analysis was conducted using Stata 17.0 software. The primary outcome variables were categorical, and effect sizes were quantified as odds ratio (OR), with 95% confidence intervals (CI). Heterogeneity among study results was assessed using Cochran's Q test and the I^2 statistic, with I^2 >50% and $P<0.05$ indicating significant heterogeneity. In such cases, a random-effects model was applied; otherwise, a fixed-effect model was used. Forest plots were constructed to visualize the results. Subgroup analysis was conducted to ascertain whether the effectiveness of exercise interventions varied by exercise type, and the results were presented as forest plots. Sensitivity analysis was performed using the one-by-one elimination method to evaluate the robustness of the findings. Publication bias was assessed using a funnel plot and Egger test, with a significance level of $P<0.05$. Furthermore, a network meta-analysis was conducted, involving the construction of an evidence network, calculation of the surface under the cumulative ranking curve (SUCRA), and comparison of the efficacy of different interventions. Cumulative probability plots were created based on the SUCRA ranking, where smaller areas under the curve indicated a more favorable intervention effect.

Results

Search result

A total of 9,073 articles were initially identified through the preliminary search, and 210 articles were subsequently evaluated based on their title and abstract. Following a full-text review, 25 RCTs met the inclusion criteria. The detailed process of literature screening is illustrated in Fig. 1.

Description of included studies

A total of 25 [9–17, 20–35] RCTs were included, involving 6,824 participants. Of these, 3,389 were assigned to the experimental group and 3,435 to the control group. The studies were published between 2010 and 2021, comprising 4 in Chinese [31–34] and 21 in English [9–17, 20–30, 35]. The studies encompassed 5 distinct intervention categories: yoga (2 studies) [9, 10], aerobic exercise (2 studies) [11, 25], aerobic combined with resistance exercise (8 studies) [12, 15, 16, 22–24, 26, 27], exercise guidance (2 studies) [20, 35], and diet and exercise guidance (11 studies) [13, 14, 17, 21, 28–34]. Detailed results are presented in Additional file 2.

Risk of bias

The results of the risk of bias assessment are summarized as percentages in Fig. 2, while detailed judgments for each risk of bias item in the included studies are shown in Fig. 3. In the randomization process, all studies adopted randomization methods. However, only 7 articles explicitly mentioned allocation concealment, which was rated as low risk. The remaining 18 articles were categorized as having an unclear risk due to insufficient information. All studies were assessed as low risk in the domains of deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. Therefore, 7 studies were rated as having a low risk of overall bias, while 18 were rated as having an unclear risk.

Meta-analysis

Meta-analysis results

A total of 25 articles, encompassing 6,824 participants, reported the occurrence of GDM. The 25 studies exhibited moderate heterogeneity (I^2 = 50%, $P=0.003$). Using a random-effects model, the meta-analysis revealed that the risk of GDM was significantly lower in the exercise intervention group compared to the conventional care group (OR = 0.56, 95% CI = 0.44–0.71, $P<0.001$), as illustrated in Fig. 4.

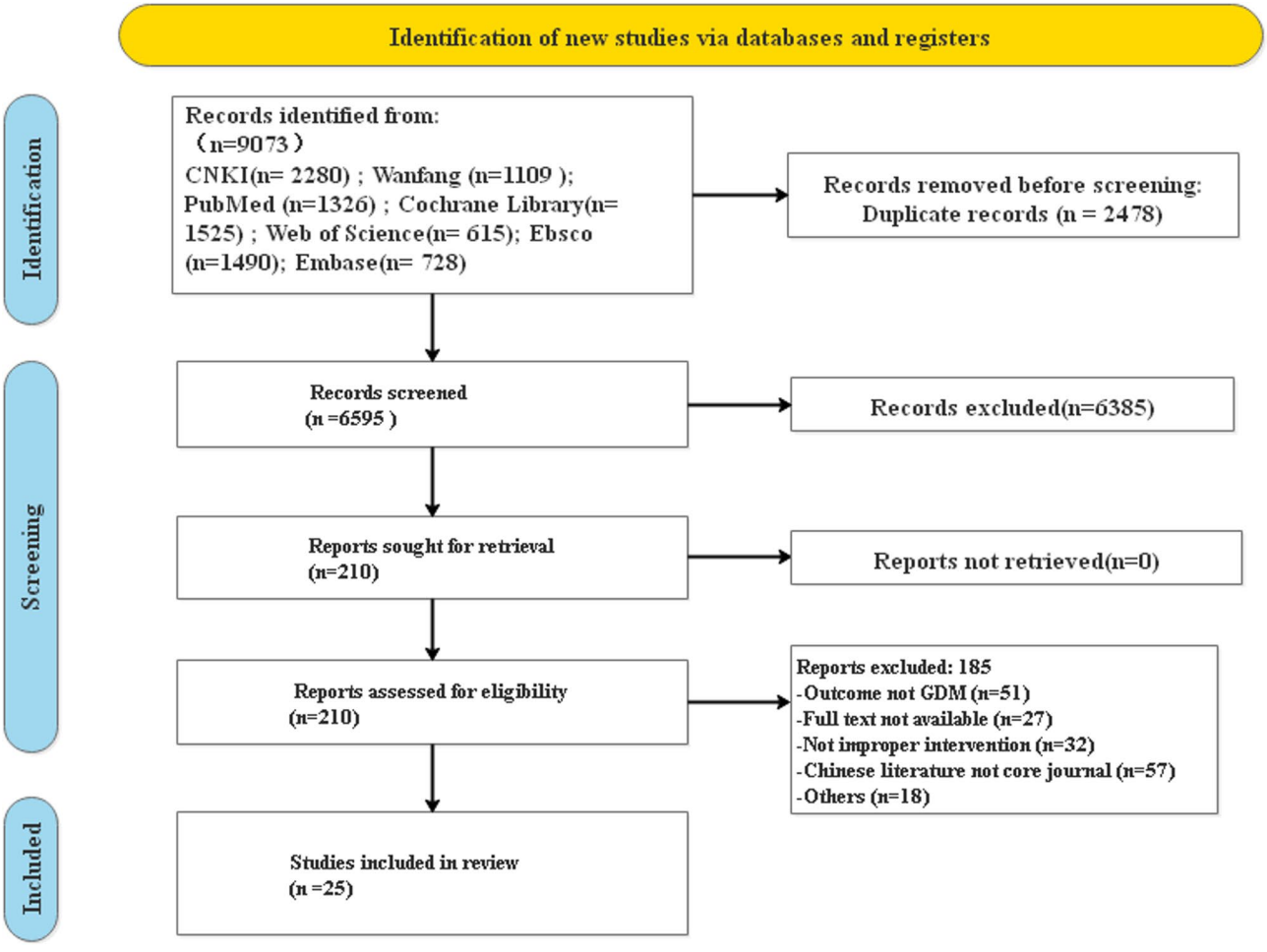

Fig. 1 Literature screening flowchart for the network meta-analysis

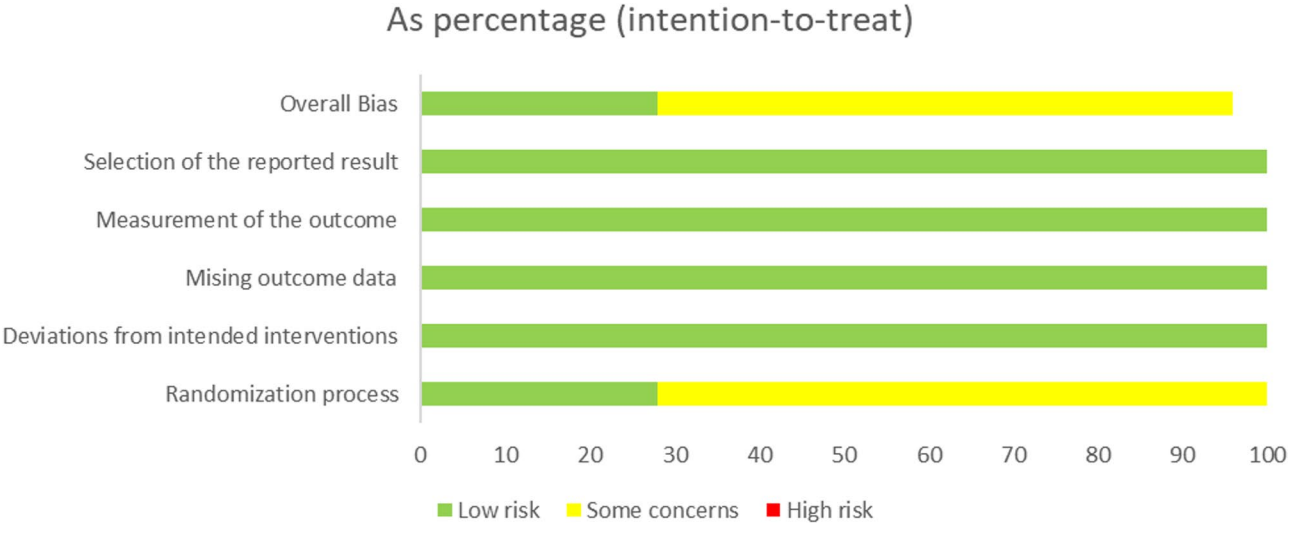

Fig. 2 Risk of bias assessment (expressed as percentages)

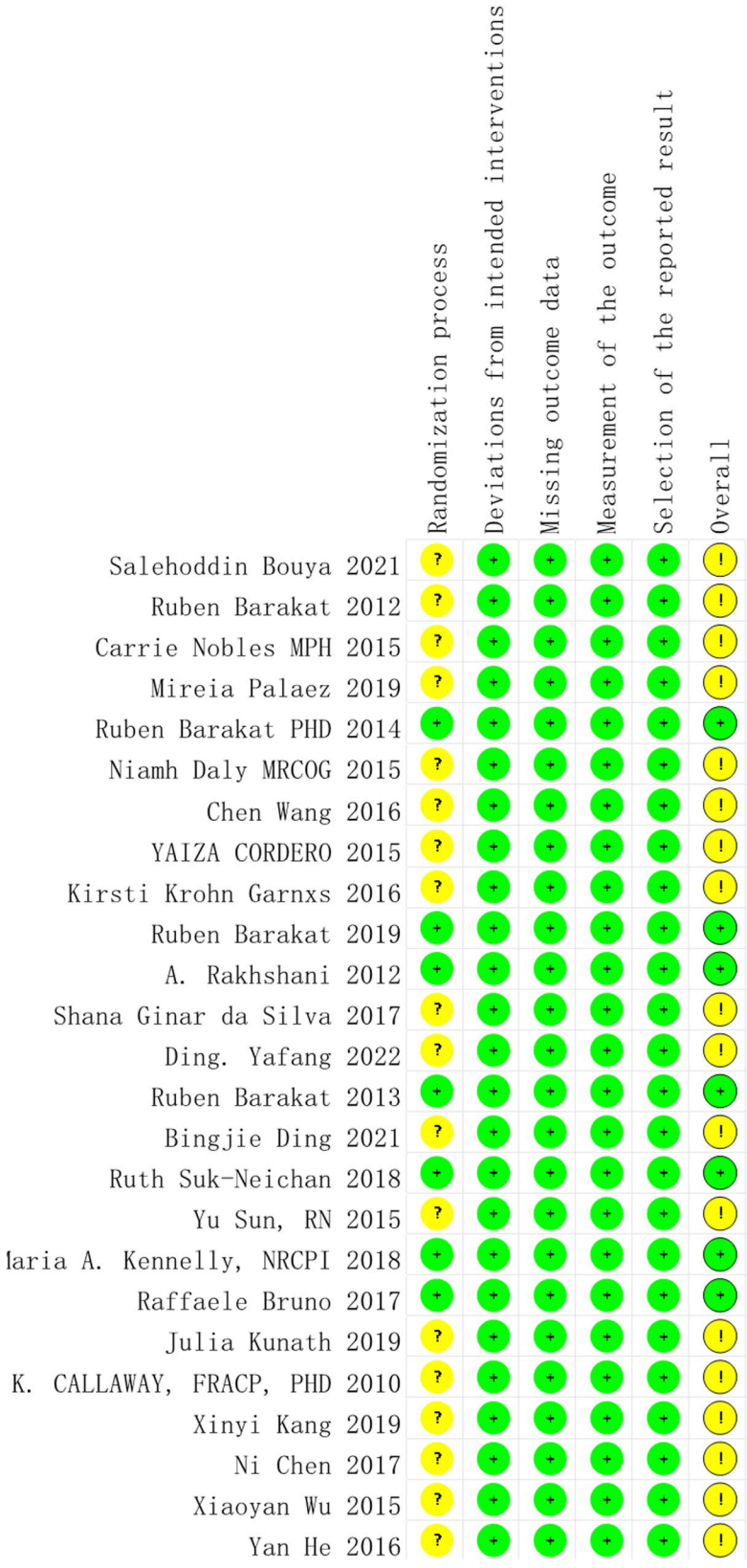

Fig. 3 The review author's judgments about each risk of bias item for the included studies

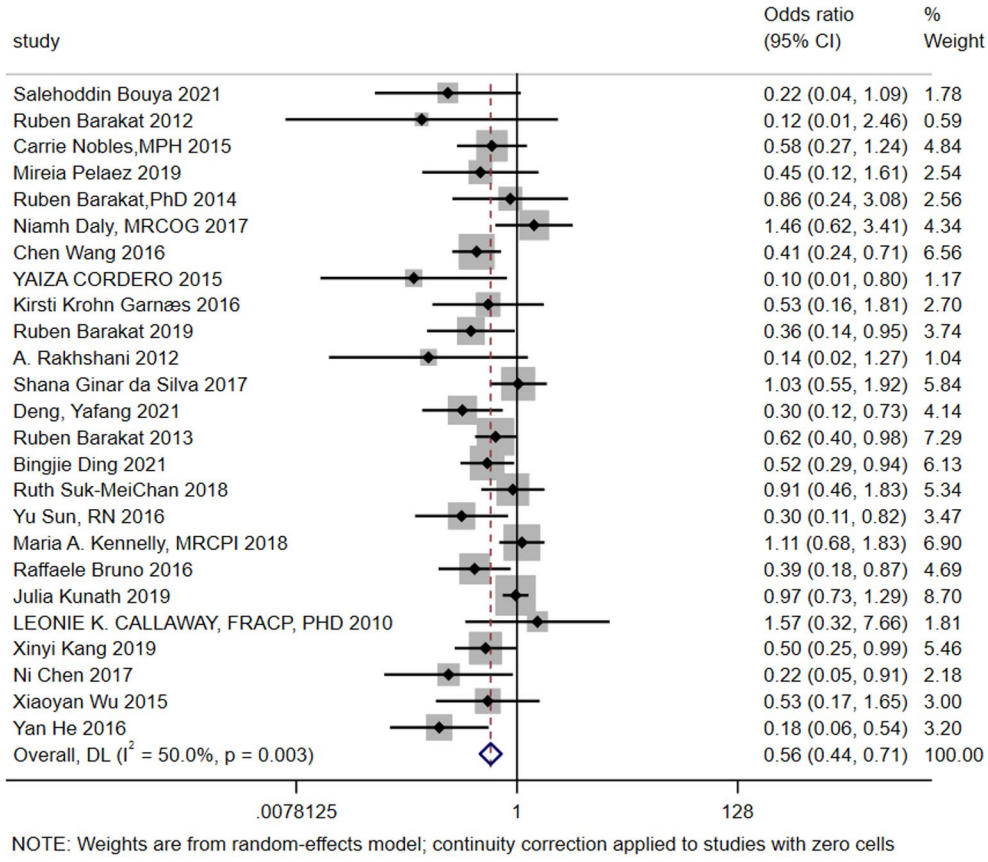

Fig. 4 Forest plot of meta-analysis

Subgroup analysis results

Subgroup analysis was performed on the types of exercise intervention Subgroup analysis was performed according to the types of exercise interventions, including yoga (I^2 = 0.0%, P = 0.758), aerobic exercise (I^2 = 0.0%, P = 0.823), aerobic combined with resistance exercise (I^2 = 29.0%, P = 0.197), exercise guidance (I^2 = 19.2%, P = 0.266), and diet and exercise guidance (I^2 = 64.1%, P = 0.002). The results demonstrated that, compared to the conventional care group, aerobic exercise (OR = 0.40, 95% CI = 0.25–0.64, P < 0.001), yoga (OR = 0.19, 95% CI = 0.05–0.69, P = 0.011), and diet and exercise guidance (OR = 0.53, 95% CI = 0.37–0.76, P = 0.001) significantly reduced the risk of GDM. Conversely, aerobic combined with resistance exercise (OR = 0.71, 95% CI = 0.47–1.08, P = 0.107) and exercise guidance (OR = 0.74, 95% CI = 0.32–1.73, P = 0.488) exhibited no statistically significant effect, as illustrated in Fig. 5.

Subgroup analysis by population characteristics Subgroup analysis was conducted according to the population characteristics, classifying participants into three categories: healthy individuals (I^2 = 45.6%, P = 0.065), overweight and/or obese individuals (I^2 = 56.7%, P = 0.010), and pregnant women at risk of GDM (I^2 = 25.2%, P = 0.253). The results indicated that exercise intervention during pregnancy significantly reduced the risk of GDM across all subgroups: healthy individuals (OR = 0.63, 95% CI = 0.43–0.93, P = 0.019), overweight and/or obese individuals (OR = 0.53, 95% CI = 0.36–0.79, P = 0.002), and pregnant women at risk of GDM (OR = 0.53, 95% CI = 0.34–0.82, P < 0.004), as illustrated in Fig. 6.

Subgroup analysis by intervention initiation time Subgroup analysis was performed according to the timing of intervention initiation, categorized as early pregnancy (I^2 = 45.3%, P = 0.020), middle pregnancy (I^2 = 67.7%, P = 0.079), and early and middle pregnancy (I^2 = 69.4%, P = 0.020). The results demonstrated that initiating exercise interventions in early pregnancy significantly reduced the risk of GDM (OR = 0.51, 95% CI = 0.39–0.68, P < 0.001). In contrast, interventions initiated during the second trimester (OR = 0.57, 95% CI = 0.13–2.48, P = 0.453) or span-

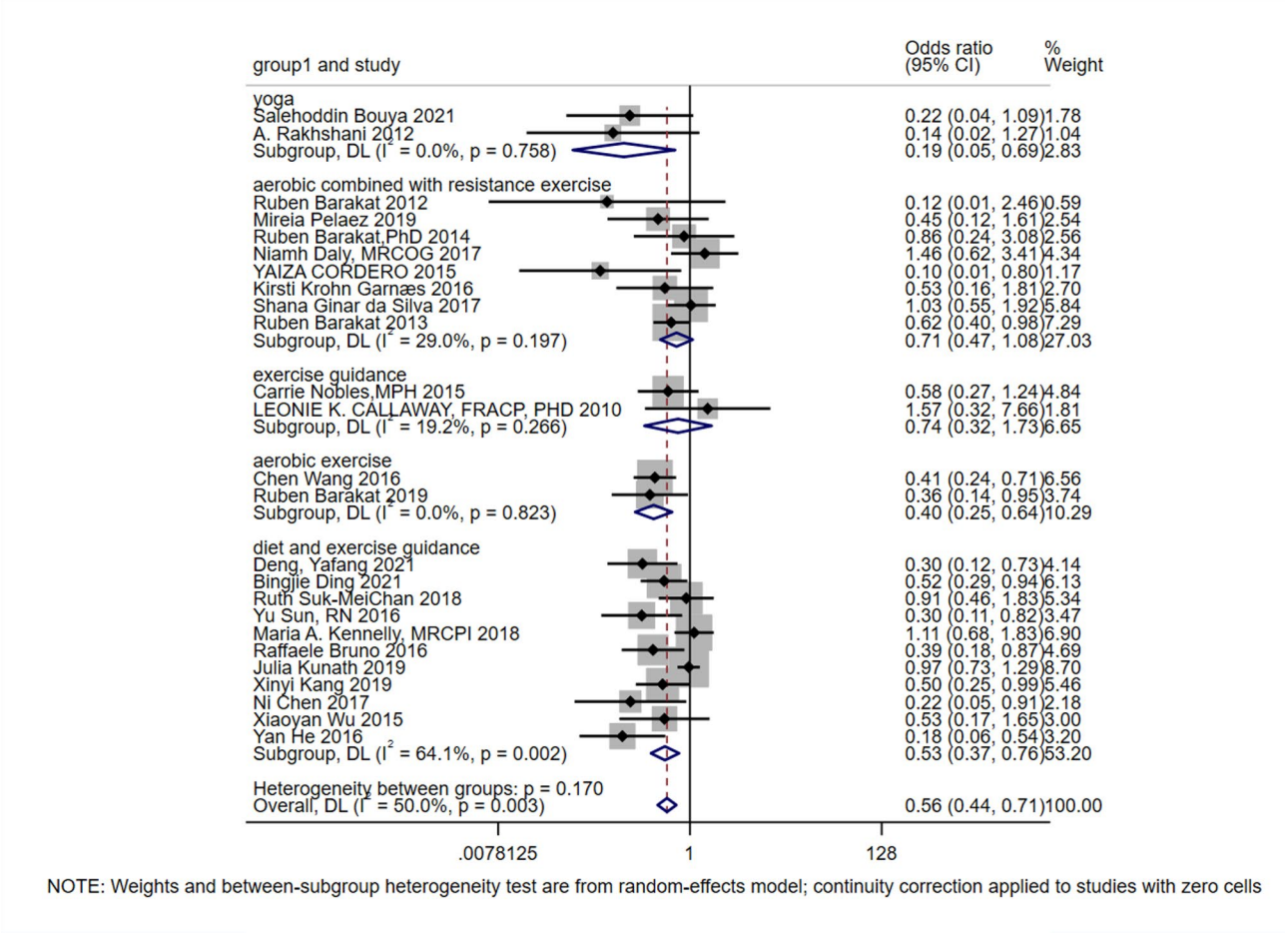

Fig. 5 Forest plot of subgroup analysis about the types of exercise interventions

ning both early and middle pregnancy (OR = 0.78, 95% CI = 0.41–1.48, $P = 0.446$) did not yield statistically significant effects, as illustrated in Fig. 7.

Sensitivity analysis

Sensitivity analysis was conducted using the one-by-one elimination method, in which each study was sequentially excluded, and the remaining articles (n-1) were re-analyzed. The purpose was to assess the influence of individual studies on the overall meta-analysis results. As shown in Fig. 8, the combined results of the remaining studies remained statistically significant (95% CI < 1) after the exclusion of most studies. This indicates that the original meta-analysis results were stable and not substantially affected by changes in the number of included studies, demonstrating their robustness.

Publication bias

The funnel plot (Fig. 9) and the Egger test results ($p = 0.002$) indicate the potential presence of publication bias among the included RCTs. A trim-and-fill analysis was conducted, adding six hypothetical studies to account for the bias. The adjusted funnel plot is presented in Fig. 10 (OR = 0.63, 95% CI = 0.50–0.81). The results before and after the trim-and-fill adjustment remained statistically significant, confirming the stability of the findings.

Network meta-analysis results

Network plot

The prevalence of GDM was reported in 25 studies that encompassed five distinct exercise interventions: aerobic exercise, aerobic combined with resistance exercise, yoga, exercise guidance, and diet and exercise guidance. The control group was treated with conventional care. The distribution of studies across the interventions was as follows: 2 studies on yoga, 2 on aerobic exercise, 8 on aerobic combined with resistance exercise, 2 on exercise guidance, and 11 articles on diet and exercise guidance. The network evidence graph, illustrating the relationships between these interventions, is visually represented in Fig. 11.

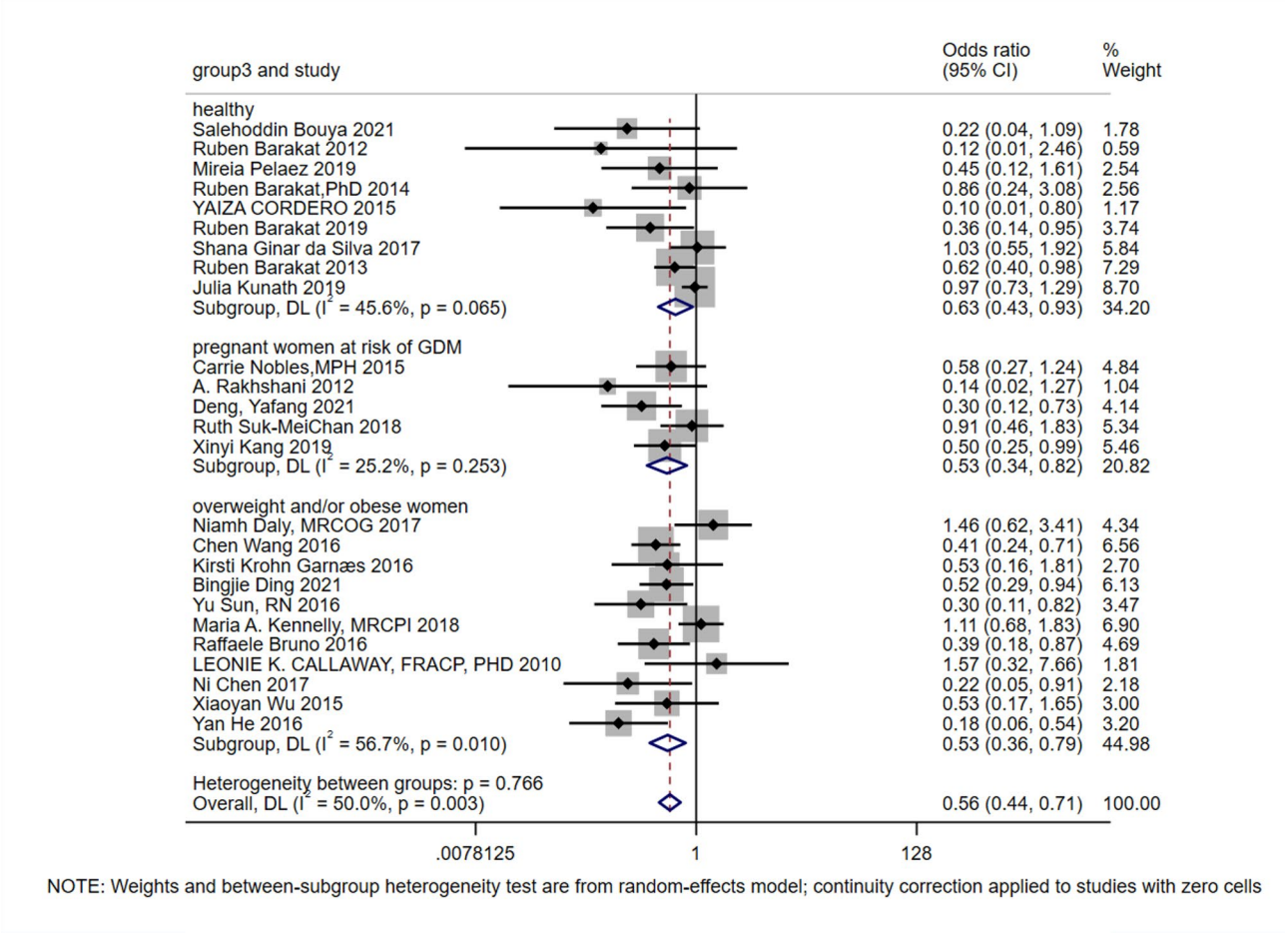

Fig. 6 Forest plot of subgroup analysis by population characteristics

Network meta-analysis

The findings indicated that compared to conventional care, aerobic exercise, yoga, and diet and exercise guidance significantly reduced the risk of GDM ($P < 0.05$). Conversely, other interventions showed no significant differences in reducing GDM. These findings are detailed in Table 3; Fig. 12.

SUCRA ranking

The SUCRA rankings for all interventions and control measures are presented in Fig. 13. A smaller SUCRA value indicates a lower risk of GDM and greater effectiveness of the intervention. Based on SUCRA rankings, yoga emerged as the most effective intervention for preventing GDM, with the lowest SUCRA value (6.8%). It was followed by aerobic exercise (SUCRA = 25.4%), diet and exercise guidance (SUCRA = 43.5%), aerobic combined with resistance exercise (SUCRA = 63.4%), exercise guidance (SUCRA = 68.0%) and conventional care (SUCRA = 92.9%). These findings confirm that yoga is the most efficacious exercise intervention for reducing the risk of GDM.

Discussion

GDM is a prevalent complication during pregnancy that poses serious health risks to both the mother and fetus. Physical activity during pregnancy has been identified as an effective strategy for promoting maternal health. Prior research has demonstrated that exercise not only helps regulate blood glucose levels in patients [36] but also serves as an effective method for the prevention of GDM [37]. Furthermore, compared to other intervention methods, exercise offers several benefits, including safety, efficacy, and minimal risk of adverse effects, making it particularly suitable for pregnant women. Consequently, providing appropriate exercise recommendations for pregnant women is crucial to maintaining health and preventing complications during pregnancy.

The findings of the meta-analysis indicate that exercise is an effective method for the prevention of GDM, consistent with prior research findings [37]. However, the precise mechanisms by which exercise prevents GDM remain an active area of investigation. Current evidence suggests that the etiology of GDM shares similarities with that of T2DM, with insulin resistance being a key

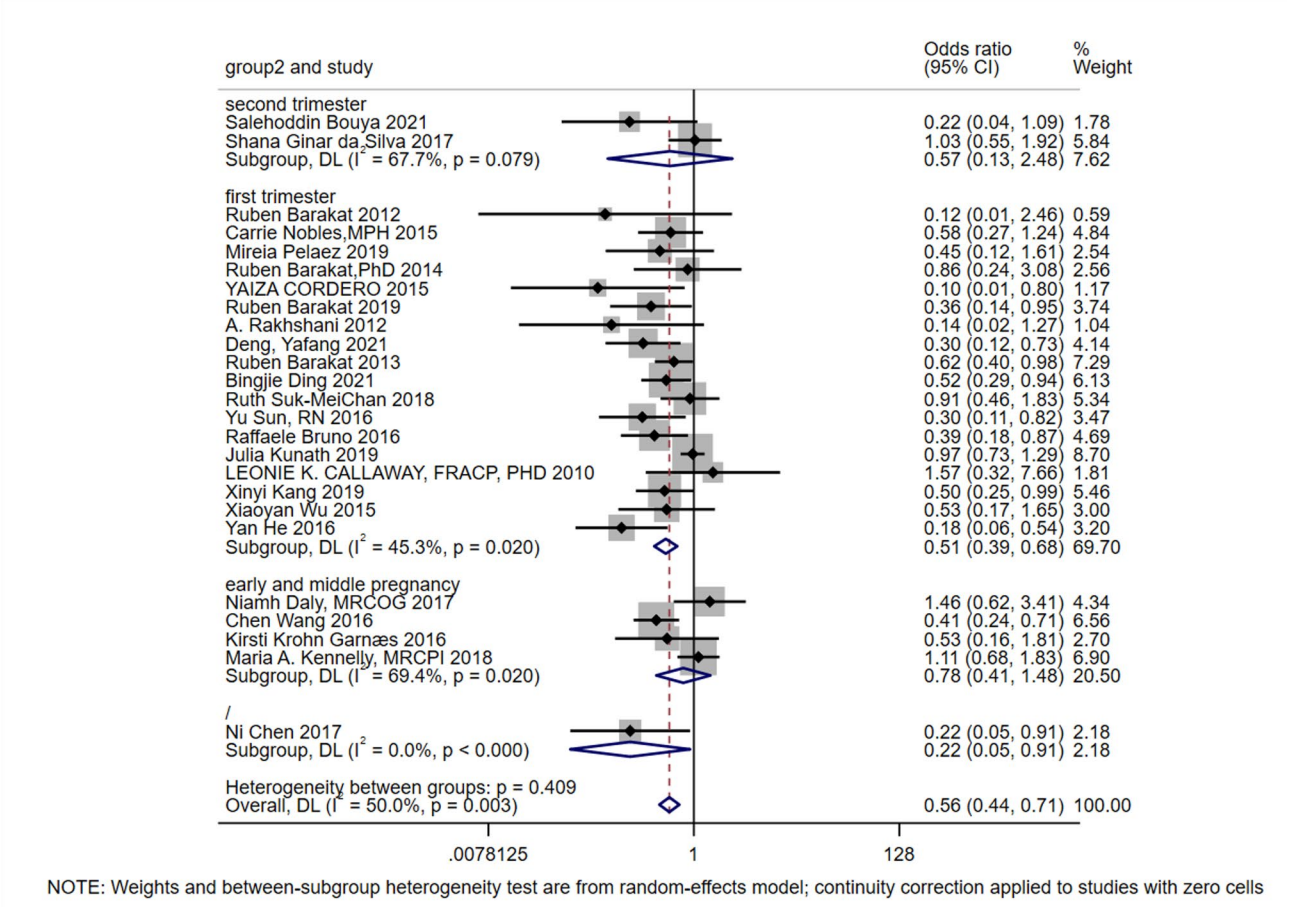

Fig. 7 Forest plot of subgroup analysis by intervention initiation time

underlying factor. Exercise has been shown to reduce insulin resistance, thereby lowering the risk of GDM [38]. The mechanisms by which exercise mitigates insulin resistance are multifaceted. Firstly, studies have demonstrated that exercise can compensate for the defects of the insulin signaling pathway, offering an alternative method to enhance glucose uptake and facilitate insulin transport [38–40]. Secondly, exercise has been shown to reduce insulin resistance by altering the distribution of specific adipokines [38, 41, 42]. Thirdly, exercise has been shown to reduce inflammation, decrease free radical levels, lower oxidative stress, and thereby protect the function of islet β cells [38, 43, 44].

Subgroup analysis revealed that compared to the conventional care group, aerobic exercise, yoga and diet and exercise guidance were significantly associated with a notably reduced risk of GDM. Aerobic exercise, which relies on aerobic metabolism to generate energy, includes activities such as brisk walking, jogging, swimming, and cycling. It has been shown to enhance glucose uptake and utilization, improve metabolic function, regulate blood sugar levels, and lower the risk of GDM [45, 46]. Yoga a holistic practice, has been shown to enhance physical, psychological, and mental well-being [47]. Research indicates that yoga can alleviate stress in pregnant women [10], and its meditation and relaxation components may contribute to GDM prevention [9]. Diet and exercise guidance also played a key role in GDM prevention by positively influencing maternal metabolism, glycemic control, weight management, and overall health [48]. Improved dietary habits and physical activity can enhance insulin sensitivity, regulate blood sugar, and lower body mass index (BMI) — a major risk factor for GDM in pregnant women [49]. However, in this study, exercise guidance and aerobics combined with resistance exercise did not show a significant effect in reducing the risk of GDM. The lack of preventive efficacy from exercise guidance may be attributed to its reliance on advice alone, without professional guidance or supervision, leading to low compliance and limited exercise effectiveness [20, 35]. Similarly, the ineffectiveness of aerobic combined with resistance exercise might be due to variations in the initiation time and duration of the interventions across studies. Additionally, the higher intensity

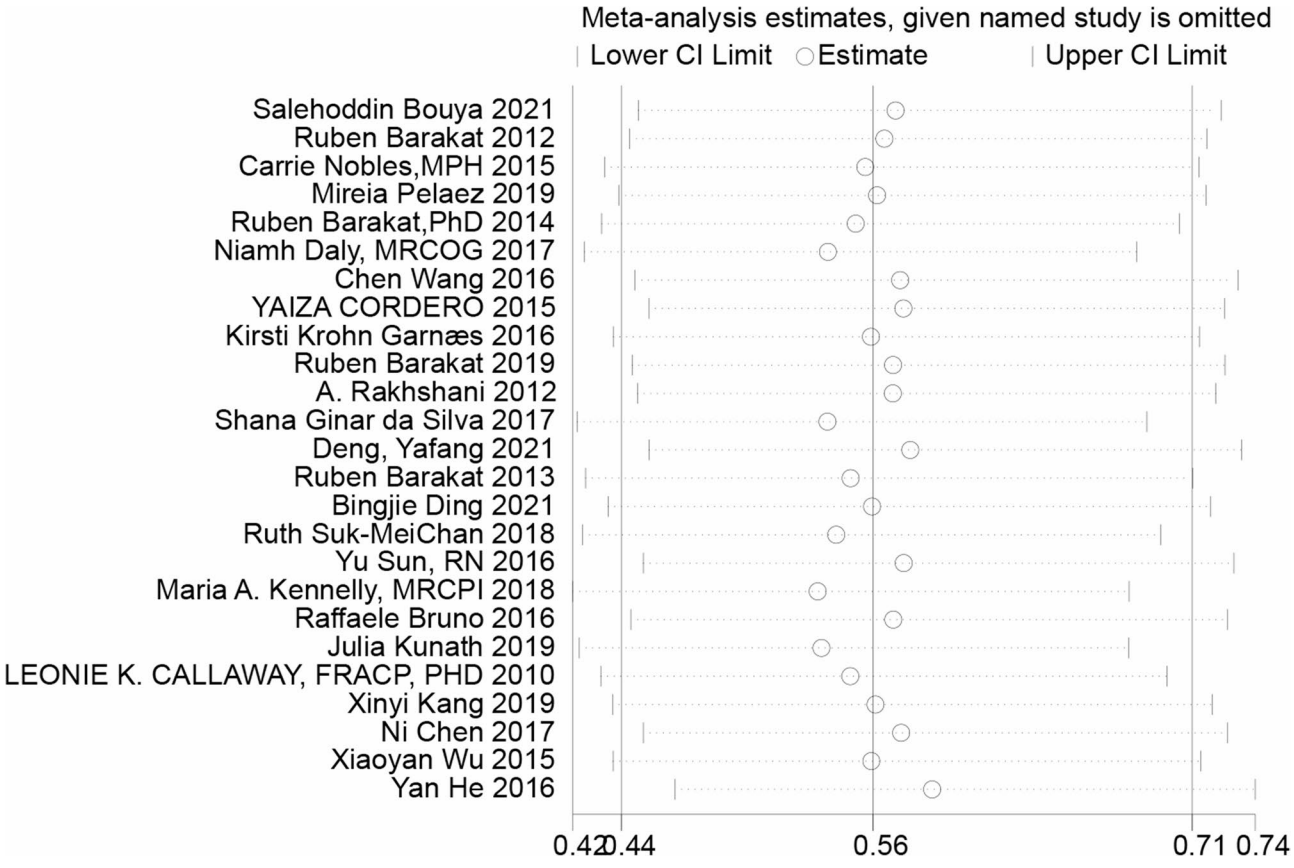

Fig. 8 The plot of sensitivity analysis

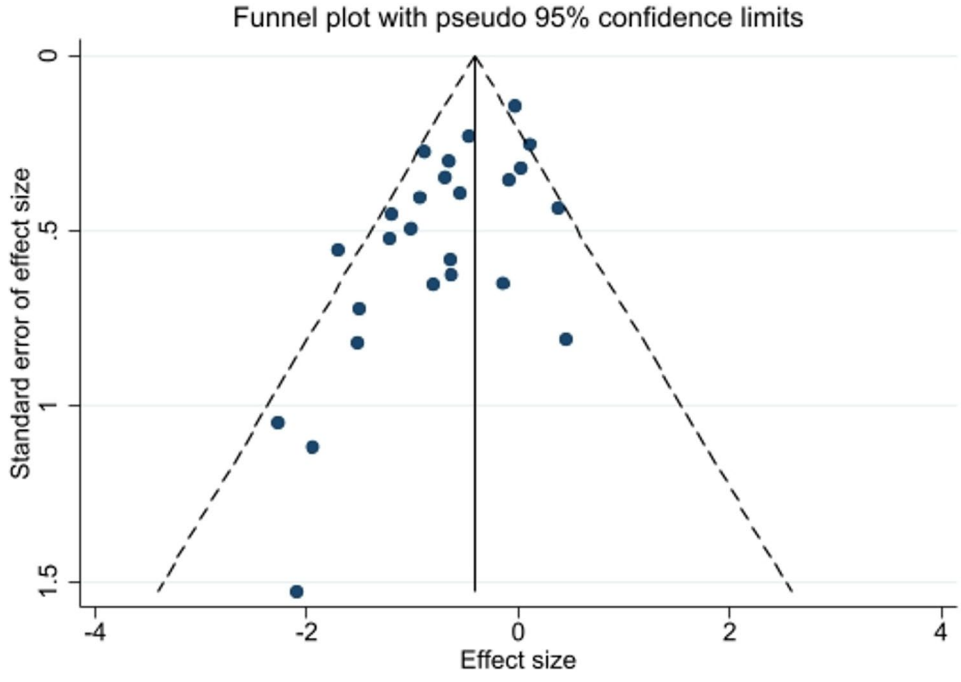

Fig. 9 The funnel plot of meta-analysis

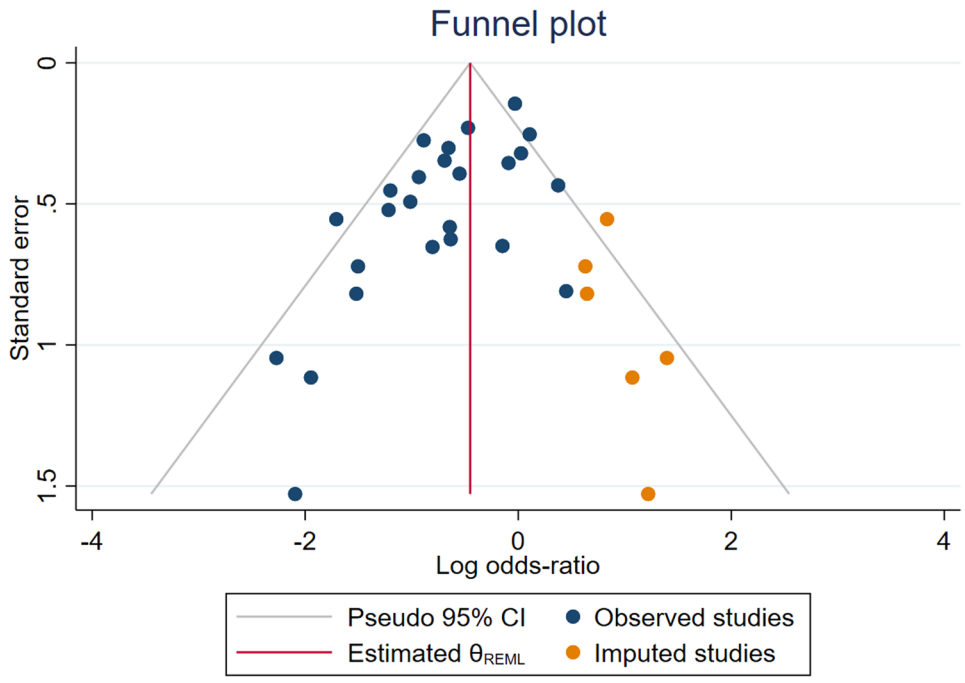

Fig. 10 The funnel plot after trim-and-fill analysis

and potential risks associated with resistance exercises may render them less suitable for pregnant individuals, thereby limiting the overall quality and impact of these interventions [12, 15, 16, 22–24, 26, 27]. Furthermore, subgroup analysis was conducted to examine both the characteristics of the intervention population and the timing of the intervention. The findings indicated that exercise interventions during pregnancy effectively prevented the occurrence of GDM in healthy individuals, overweight and obese individuals, and pregnant women at risk of GDM. Additionally, exercise intervention initiated in early pregnancy was also effective in preventing the onset of GDM. These results are consistent with previous studies [37].

The results of the network meta-analysis indicate that yoga may be the most efficacious exercise intervention for preventing GDM. Yoga incorporates millennia-old techniques that are relatively simple to learn and aim to enhance physical, psychological, emotional, and spiritual well-being. These techniques include postures to purify the body, breathing exercises to cleanse the breath, and meditative practices to calm the mind [50]. Pregnancy is a period of heightened physical and emotional stress for women, particularly due to the cardiovascular strain it places on the body [51]. Yoga has gained popularity for its ability to reduce blood pressure and alleviate stress, making it a particularly suitable exercise for pregnant women [52, 53]. Although the precise mechanism by which yoga prevents GDM remains unclear, its unique advantages set it apart from other forms of exercise. Firstly, yoga is especially well-suited for pregnant women [9] as it requires no complex equipment or specialized location. Its gentle movements and slow rhythm reduce the risk associated with high-intensity exercises, allowing women to safely practice yoga throughout pregnancy. Secondly, yoga extends beyond physical activity to include breathing and meditation, which benefit not only physical fitness but also the emotional and psychological well-being of pregnant women. Finally, yoga supports the regulation of the endocrine system, helping to balance hormone secretion [54]. This hormonal balance may contribute to stabilizing blood glucose levels, thereby reducing the risk of GDM.

Previous systematic reviews and meta-analyses have explored the preventive effects of exercise on GDM. For instance, Shan Wu et al. [55] focused on overweight and obese pregnant women, comparing the effects of diet, physical activity, and combined interventions. The results indicated that although neither dietary intervention, physical activity alone, nor their combination effectively reduced the risk of GDM in this population, there was a trend suggesting that physical activity combined with diet could be a preventive factor for GDM. Similarly, Tsironikos et al. [56] conducted a systematic review of high-risk pregnant women and found that lifestyle interventions during pregnancy may be beneficial in reducing the incidence of GDM. Another meta-analysis by Vicente Martínez-Vizcaíno et al. [37] demonstrated that exercise significantly lowers the risk of GDM in pregnant women who are not overweight or obese. Notably, the benefits were most pronounced when interventions were delivered by trained professionals, involved low to moderate intensity, and were initiated within the first

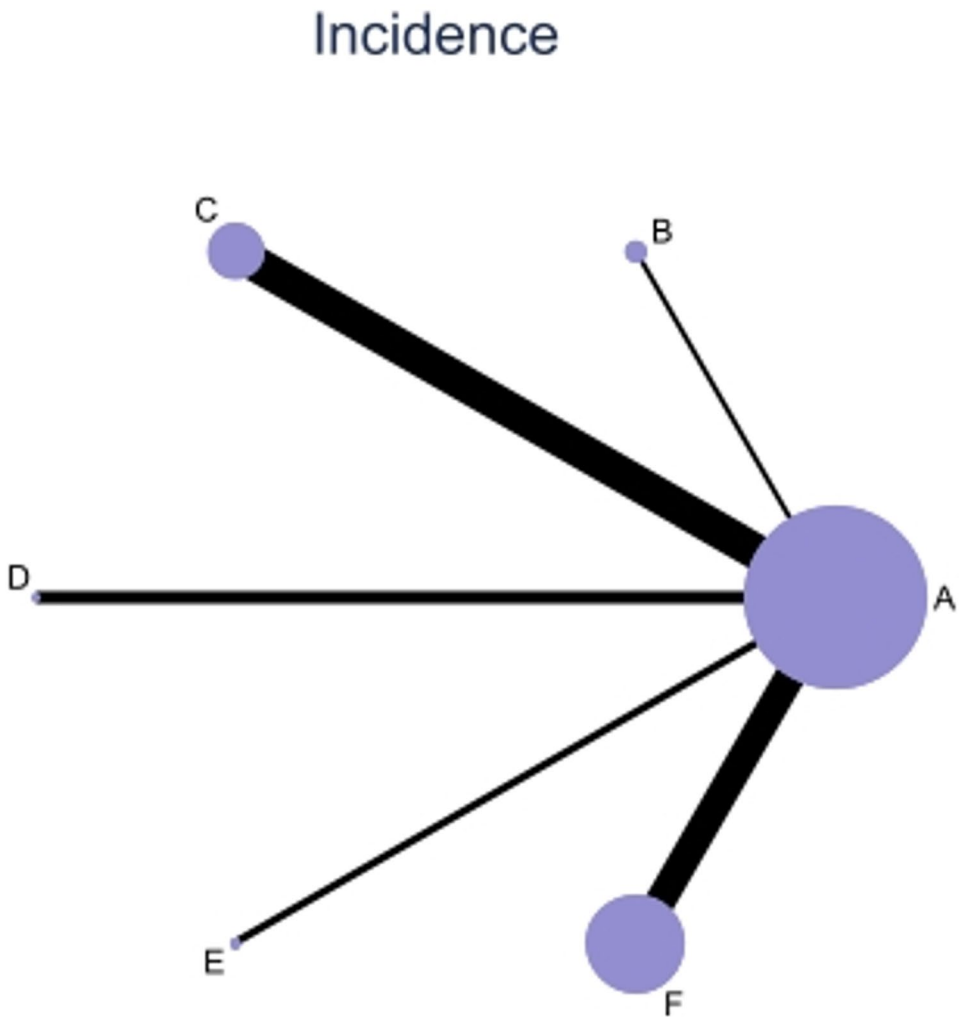

Fig. 11 The plot of evidence network. Note: A = conventional care; B = aerobic exercise; C = aerobic combined with resistance exercise; D = Yoga; E = exercise guidance; F = diet and exercise guidance

Table 3 Network meta-analysis of the preventive effects of different interventions on GDM (OR and 95% CI)

A	**0.39 (0.19,0.82)**	**0.70 (0.44,1.11)**	**0.19 (0.05,0.76)**	**0.76 (0.30,1.89)**	**0.55 (0.39,0.78)**
2.54 (1.22,5.28)	B	1.78 (0.75,4.21)	0.47 (0.10,2.31)	1.92 (0.59,6.21)	1.40 (0.62,3.13)
1.43 (0.90,2.26)	0.56 (0.24,1.33)	C	0.27 (0.06,1.17)	1.08 (0.39,3.02)	0.79 (0.45,1.37)
5.35 (1.32,21.76)	2.11 (0.43,10.25)	3.75 (0.86,16.37)	D	4.05 (0.76,21.63)	2.94 (0.70,12.47)
1.32 (0.53,3.30)	0.52 (0.16,1.68)	0.93 (0.33,2.59)	0.25 (0.05,1.32)	E	0.73 (0.27,1.95)
1.82 (1.29,2.56)	0.72 (0.32,1.60)	1.27 (0.73,2.22)	0.34 (0.08,1.44)	1.37 (0.51,3.69)	F

Note: A = conventional care; B = aerobic exercise; C = aerobic combined with resistance exercise; D = Yoga; E = exercise guidance; F = diet and exercise guidance

trimester. While these studies provide valuable insights into the preventive role of exercise for GDM, their conclusions are primarily limited to general efficacy or specific subpopulations. In addition, our previous systematic review demonstrated a clear dose–response relationship between physical activity and GDM risk, with the lowest risk observed at approximately 50 MET-hours per week [6]. This finding provided quantitative evidence on an approximate activity volume associated with the greatest preventive benefit. Extending these findings, the present study addresses the critical question of which type of exercise is most effective. By employing a network meta-analysis that included a diverse range of participants, we comprehensively compared the preventive effects of various exercise modalities. A key finding of this study is that yoga may be the most effective intervention for preventing GDM. Taken together, these two reviews offer complementary insights, providing both quantitative targets and practical recommendations for clinical and public health practice.

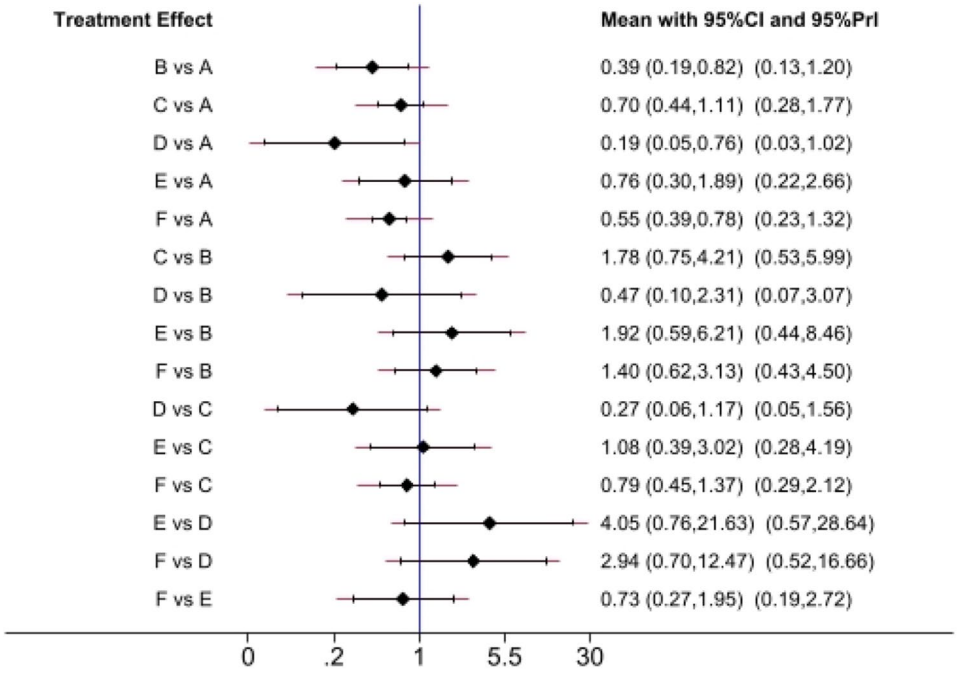

Fig. 12 Forest plot of network meta-analysis. Note: A = conventional care; B = aerobic exercise; C = aerobic combined with resistance exercise; D = Yoga; E = exercise guidance; F = diet and exercise guidance

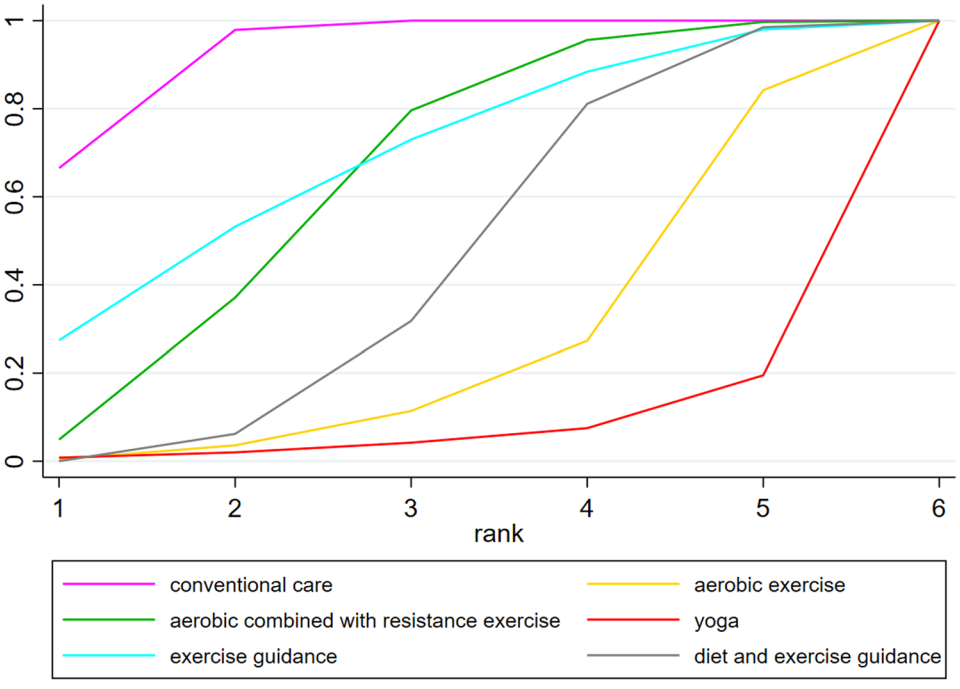

Fig. 13 SUCRA ranking of GDM risk for the different interventions

Strengths and limitation

The network meta-analysis presents several strengths: (1) This network meta-analysis systematically explored the preventive effects of five exercise interventions on GDM through a comprehensive search across seven databases. (2) This study was conducted following the PRISMA guidelines, ensuring methodological rigor. The findings provide a scientific basis for the prevention and reduction of GDM. (3) Based on the analysis of the effect of exercise on GDM prevention, this study further employed the network meta-analysis to determine the most effective

exercise intervention, offering evidence for precise and targeted prevention.

This review has several noteworthy limitations: (1) The limited number of included studies, particularly those focusing on yoga and aerobic exercise, may introduce potential bias into the results. (2) Comparisons between different types of exercise were indirect, as the included studies primarily compared exercise interventions to conventional care. (3) Only Chinese and English literature were included, potentially introducing publication bias. Further high-quality, large-sample studies in diverse populations are required to validate these findings.

Conclusion

Exercise during pregnancy is an effective method for preventing GDM, particularly when initiated early. Among the interventions analyzed, yoga, aerobic exercise, and diet and exercise guidance were shown to significantly reduce the risk of GDM. Yoga, in particular, appears to be the most effective and accessible option. Its simplicity, safety, and holistic benefits make yoga a potential recommended practice for pregnant women. However, there is a clear need for high-quality, large-sample RCTs to provide more robust evidence in the future.

Abbreviations

GDM	Gestational diabetes mellitus
T2DM	Type 2 diabetes mellitus
IADPSG	International association of diabetes and pregnancy research groups
WHO	World health organization
PRISMA	Reporting items for systematic reviews and meta-analyses
PROSPERO	Prospective register of systematic reviews
RCT	Randomized controlled trial
RoB	Risk of bias
OR	Odds ratio
CI	Confidence interval
SUCRA	Surface under the cumulative ranking curve
BMI	Body mass index

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-24807-5.

Supplementary Material 1.

Supplementary Material 2.

Acknowledgements

Not applicable.

Authors' contributions

L.Z. and J.C. conceptualized and designed this study. Y.R., Y.G., and Y.H. conducted database searches, data extraction, and analysis. Y.R. wrote the manuscript. L.Z. revised the manuscript. All authors critically reviewed and approved the manuscript as submitted. L.Z. (2466@bsu.edu.cn) and J. C. (cjy19781031@163.com) are co-corresponding authors of this manuscript.

Funding

This work was supported by the National Natural Science Foundation of China [Grant number 81803324], the Food Science and Technology Fund of the Chinese Institute of Food Science and Technology - SCITOP BIO Young Scientists Fund [Grant number 2018-02], and the Chinese Universities Scientific Fund [Grant number 2024TZJK007].

Data availability

All data are available from the corresponding author on reasonable request: Liuwei Zhang [(2466@bsu.edu.cn)].

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

[1]Key Laboratory of Exercise and Physical Fitness (Beijing Sport University), Ministry of Education, Beijing, China
[2]Department of Physical Fitness and Health, School of Sport Science, Beijing Sport University, Beijing 100084, China
[3]Department of Obstetrics and Gynecology, China-Japan Friendship Hospital, Beijing 100029, China
[4]School of Strength and Conditioning, Beijing Sport University, Beijing 100084, China
[5]School of Education, Beijing Sport University, Beijing 100084, China

Received: 25 November 2024 / Accepted: 8 September 2025
Published online: 28 October 2025

References

1. 2. Classification and diagnosis of diabetes: standards of medical care in Diabetes-2021. Diabetes Care. 2021;44(Suppl 1):S15–33.
2. Szmuilowicz ED, Josefson JL, Metzger BE. Gestational diabetes mellitus. Endocrinol Metab Clin North Am. 2019;48(3):479–93.
3. Sweeting A, Hannah W, Backman H, Catalano P, Feghali M, Herman WH, Hivert MF, Immanuel J, Meek C, Oppermann ML, et al. Epidemiology and management of gestational diabetes. Lancet. 2024;404(10448):175–92.
4. Wang C, Zhu W, Wei Y, Feng H, Su R, Yang H. Exercise intervention during pregnancy can be used to manage weight gain and improve pregnancy outcomes in women with gestational diabetes mellitus. BMC Pregnancy Childbirth. 2015;15:255.
5. Bull FC, Al-Ansari SS, Biddle S, Borodulin K, Buman MP, Cardon G, Carty C, Chaput JP, Chastin S, Chou R, et al. World health organization 2020 guidelines on physical activity and sedentary behaviour. Br J Sports Med. 2020;54(24):1451–62.
6. Xie W, Zhang L, Cheng J, Wang Y, Kang H, Gao Y. Physical activity during pregnancy and the risk of gestational diabetes mellitus: a systematic review and dose-response meta-analysis. BMC Public Health. 2024;24(1):594.
7. Ribeiro MM, Andrade A, Nunes I. Physical exercise in pregnancy: benefits, risks and prescription. J Perinat Med. 2022;50(1):4–17.
8. Gallo-Galán LM, Gallo-Vallejo MA, Gallo-Vallejo JL. [Physical exercise and pregnancy. Evidence based medicine (EBM)]. Semergen. 2022;48(6):423–30.
9. Bouya S, Rezaie Keikhaie L, Hosseini S, Rezaie Keikhaie K. The effect of yoga on uterine artery doppler indices, maternal and fetal complications in pregnant women: A quasi-experimental study. J Ayurveda Integr Med. 2021;12(1):70–4.
10. Rakhshani A, Nagarathna R, Mhaskar R, Mhaskar A, Thomas A, Gunasheela S. The effects of yoga in prevention of pregnancy complications in high-risk pregnancies: A randomized controlled trial. Prev Med. 2012;55(4):333–40.
11. Wang C, Wei Y, Zhang X, Zhang Y, Xu Q, Su S, Zhang L, Liu C, Feng Y, Shou C, et al. Effect of regular exercise commenced in early pregnancy on the incidence of gestational diabetes mellitus in overweight and obese pregnant women: A randomized controlled trial. Diabetes Care. 2016;39(10):e163–4.
12. Cordero Y, Mottola MF, Vargas J, Blanco M, Barakat R. Exercise is associated with a reduction in gestational diabetes mellitus. Med Sci Sports Exerc. 2015;47(7):1328–33.

13. Ding B, Gou B, Guan H, Wang J, Bi Y, Hong Z. WeChat-assisted dietary and exercise intervention for prevention of gestational diabetes mellitus in overweight/obese pregnant women: a two-arm randomized clinical trial. Arch Gynecol Obstet. 2021;304(3):609–18.
14. Deng Y, Hou Y, Wu L, Liu Y, Ma L, Yao A. Effects of diet and exercise interventions to prevent gestational diabetes mellitus in pregnant women with High-Risk factors in china: A randomized controlled study. Clin Nurs Res. 2021;31(5):836–47.
15. Barakat R, Cordero Y, Coteron J, Luaces M, Montejo R. Exercise during pregnancy improves maternal glucose screen at 24–28 weeks: a randomised controlled trial. Br J Sports Med. 2012;46(9):656–61.
16. Pelaez M, Gonzalez-Cerron S, Montejo R, Barakat R. Protective effect of exercise in pregnant women including those who exceed weight gain recommendations: A randomized controlled trial. Mayo Clin Proc. 2019;94(10):1951–9.
17. Chan RS-M, Tam W-H, Ho IC-H, Kwan MW-C, Li LS, Sea MM-M, Woo J. Randomized trial examining effectiveness of lifestyle intervention in reducing gestational diabetes in high risk Chinese pregnant women in Hong Kong. Sci Rep. 2018;8(1):13849.
18. Wang C, Wei Y, Zhang X, Zhang Y, Xu Q, Sun Y, Su S, Zhang L, Liu C, Feng Y, et al. A randomized clinical trial of exercise during pregnancy to prevent gestational diabetes mellitus and improve pregnancy outcome in overweight and obese pregnant women. Am J Obstet Gynecol. 2017;216(4):340–51.
19. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 2021;372:n71.
20. Nobles C, Marcus BH, Stanek EJ, Braun B, Whitcomb BW, Solomon CG, Manson JE, Markenson G, Chasan-Taber L. Effect of an exercise intervention on gestational diabetes mellitus. Obstet Gynecol. 2015;125(5):1195–204.
21. Kunath J, Günther J, Rauh K, Hoffmann J, Stecher L, Rosenfeld E, Kick L, Ulm K, Hauner H. Effects of a lifestyle intervention during pregnancy to prevent excessive gestational weight gain in routine care – the cluster-randomised GeliS trial. BMC Med 2019;17(1):5.
22. Barakat R, Perales M, Bacchi M, Coteron J, Refoyo I. A program of exercise throughout Pregnancy. Is it safe to mother and newborn? Am J Health Promotion. 2014;29(1):2–8.
23. Daly N, Farren M, McKeating A, O'Kelly R, Stapleton M, Turner MJ. A medically supervised pregnancy exercise intervention in obese women. Obstet Gynecol. 2017;130(5):1001–10.
24. Wareham NJ, Garnæs KK, Mørkved S, Salvesen Ø, Moholdt T. Exercise training and weight gain in obese pregnant women: A randomized controlled trial (ETIP trial). PLoS Med. 2016;13(7):e1002079.
25. Barakat R, Refoyo I, Coteron J, Franco E. Exercise during pregnancy has a preventative effect on excessive maternal weight gain and gestational diabetes. A randomized controlled trial. Braz J Phys Ther. 2019;23(2):148–55.
26. da Silva SG, Hallal PC, Domingues MR, Bertoldi AD, Silveira MFd, Bassani D, da Silva ICM, da Silva BGC, Coll CVN, Evenson K. A randomized controlled trial of exercise during pregnancy on maternal and neonatal outcomes: results from the PAMELA study. Int J Behav Nutr Phys Activity. 2017;14(1):175.
27. Barakat R, Pelaez M, Lopez C, Lucia A, Ruiz JR. Exercise during pregnancy and gestational diabetes-related adverse effects: a randomised controlled trial. Br J Sports Med. 2013;47(10):630–6.
28. Sun Y, Zhao H. The effectiveness of lifestyle intervention in early pregnancy to prevent gestational diabetes mellitus in Chinese overweight and obese women: A quasi-experimental study. Appl Nurs Res. 2016;30:125–30.
29. Kennelly MA, Ainscough K, Lindsay KL, O'Sullivan E, Gibney ER, McCarthy M, Segurado R, DeVito G, Maguire O, Smith T, et al. Pregnancy exercise and nutrition with smartphone application support. Obstet Gynecol. 2018;131(5):818–26.
30. Bruno R, Petrella E, Bertarini V, Pedrielli G, Neri I, Facchinetti F. Adherence to a lifestyle programme in overweight/obese pregnant women and effect on gestational diabetes mellitus: a randomized controlled trial. Matern Child Nutr 2016;13(3):e12333.
31. Xin-yi KANG, Hao WU, Jia-wen CUI, Ming-jin ZHANG, Li-ping CHEN. Effect of early lifestyle intervention on pregnancy outcome for high-risk pregnant women with gestational diabetes mellitus. Mod Prev Med. 2019;46(16):2955–9. (Chinese.).
32. Ni CHEN, Li-Hua XE, Jie-Hua JAO, He-Hui X. Preventive effect of diet and exercise guidance on gestational diabetes mellitus in overweight and obese pregnant women. Maternal Child Health Care China. 2017;32(19):4616–8. (Chinese).
33. Wu Xiaoyan J, Jie Z, Wei L. Effect of dietary and exercise intervention on the incidence of GDM in women with pre-pregnancy obesity. Mod Clin Nurs. 2015;14(09):24–7. (Chinese).
34. He Y. Analysis of the effect of diet and exercise guidance intervention on the prevention of gestational diabetes mellitus in obese patients before pregnancy. Maternal Child Health Care China. 2016;31(23):4941–3. (Chinese).
35. Callaway LK, Colditz PB, Byrne NM, Lingwood BE, Rowlands IJ, Foxcroft K, McIntyre HD. Prevention of gestational diabetes: feasibility issues for an exercise intervention in obese pregnant women. Diabetes Care. 2010;33(7):1457–9.
36. Keating N, Coveney C, McAuliffe FM, Higgins MF. Aerobic or resistance exercise for improved glycaemic control and pregnancy outcomes in women with gestational diabetes mellitus: A systematic review. Int J Environ Res Public Health 2022;19(17):10791.
37. Martínez-Vizcaíno V, Sanabria-Martínez G, Fernández-Rodríguez R, Cavero-Redondo I, Pascual-Morena C, Álvarez-Bueno C, Martínez-Hortelano JA. Exercise during pregnancy for preventing gestational diabetes mellitus and hypertensive disorders: an umbrella review of randomised controlled trials and an updated meta-analysis. BJOG. 2023;130(3):264–75.
38. Wang C, Guelfi KJ, Yang HX. Exercise and its role in gestational diabetes mellitus. Chronic Dis Transl Med. 2016;2(4):208–14.
39. de Barros MC, Lopes MA, Francisco RP, Sapienza AD, Zugaib M. Resistance exercise and glycemic control in women with gestational diabetes mellitus. Am J Obstet Gynecol. 2010;203(6):e556551–556.
40. Pessin JE, Saltiel AR. Signaling pathways in insulin action: molecular targets of insulin resistance. J Clin Invest. 2000;106(2):165–9.
41. Golbidi S, Laher I. Potential mechanisms of exercise in gestational diabetes. J Nutr Metab. 2013;2013:285948.
42. Doruk M, Uğur M, Oruç AS, Demirel N, Yildiz Y. Serum adiponectin in gestational diabetes and its relation to pregnancy outcome. J Obstet Gynaecol. 2014;34(6):471–5.
43. Nergiz S, Altınkaya ÖS, Küçük M, Yüksel H, Sezer SD, Kurt Ömürlü İ, Odabaşı AR. Circulating Galanin and IL-6 concentrations in gestational diabetes mellitus. Gynecol Endocrinol. 2014;30(3):236–40.
44. Korkmazer E, Solak N. Correlation between inflammatory markers and insulin resistance in pregnancy. J Obstet Gynaecol. 2015;35(2):142–5.
45. Yaping X, Huifen Z, Meijing Z, Huibin H, Chunhong L, Fengfeng H, Jingjing W. Effects of Moderate-Intensity aerobic exercise on blood glucose levels and pregnancy outcomes in patients with gestational diabetes mellitus: A randomized controlled trial. Diabetes Ther. 2021;12(9):2585–98.
46. Jiao-Hong CHEN, Guo-Rong LYU, Yan-Yun ZHU, Ya-Ya LI. Effect of aerobic exercise on glucose and lipid metabolism and JNK expression in offsprings of Wistar rats with gestational diabetes mellitus. Maternal Child Health Care China. 2019;34(19):4547–50. (Chinese).
47. Bijlani RL. Yoga: an ancient tool in modern medicine. Natl Med J India. 2008;21(5):215–6.
48. Parrettini S, Caroli A, Torlone E. Nutrition and metabolic adaptations in physiological and complicated pregnancy: focus on obesity and gestational diabetes. Front Endocrinol (Lausanne). 2020;11:611929.
49. Song X, Wang C, Wang T, Zhang S, Qin J. Obesity and risk of gestational diabetes mellitus: A two-sample Mendelian randomization study. Diabetes Res Clin Pract. 2023;197:110561.
50. Chandler K. The emerging field of yoga therapy. Hawaii Med J. 2001;60(11):286–7.
51. Rytlewski K, Olszanecki R, Lauterbach R, Grzyb A, Kiec-Wilk B, Dembinska-Kiec A, Basta A. Effects of oral L-arginine on the pulsatility indices of umbilical artery and middle cerebral artery in preterm labor. Eur J Obstet Gynecol Reprod Biol. 2008;138(1):23–8.
52. Field T. Prenatal exercise research. Infant Behav Dev. 2012;35(3):397–407.
53. Field T. Yoga clinical research review. Complement Ther Clin Pract. 2011;17(1):1–8.
54. Kumar RP, Dhamodhini AP, Venugopal KS, Silambanan V, Shah SKM. Role of yoga in stress management and implications in major depression disorder. J Ayurveda Integr Med. 2023;14(5):100767.
55. Wu S, Jin J, Hu K-L, Wu Y, Zhang D. Prevention of gestational diabetes mellitus and gestational weight gain restriction in Overweight/Obese pregnant women: A systematic review and network Meta-Analysis. Nutrients. 2022;14(12):2383.

56. Tsironikos GI, Potamianos P, Zakynthinos GE, Tsolaki V, Tatsioni A, Bargiota A. Effectiveness of lifestyle interventions during pregnancy on preventing gestational diabetes mellitus in High-Risk women: A systematic review and Meta-Analyses of published RCTs. J Clin Med. 2023;12(22):7038.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.